

# Lab 1 – Requirements Engineering & UML Use-Case Modelling

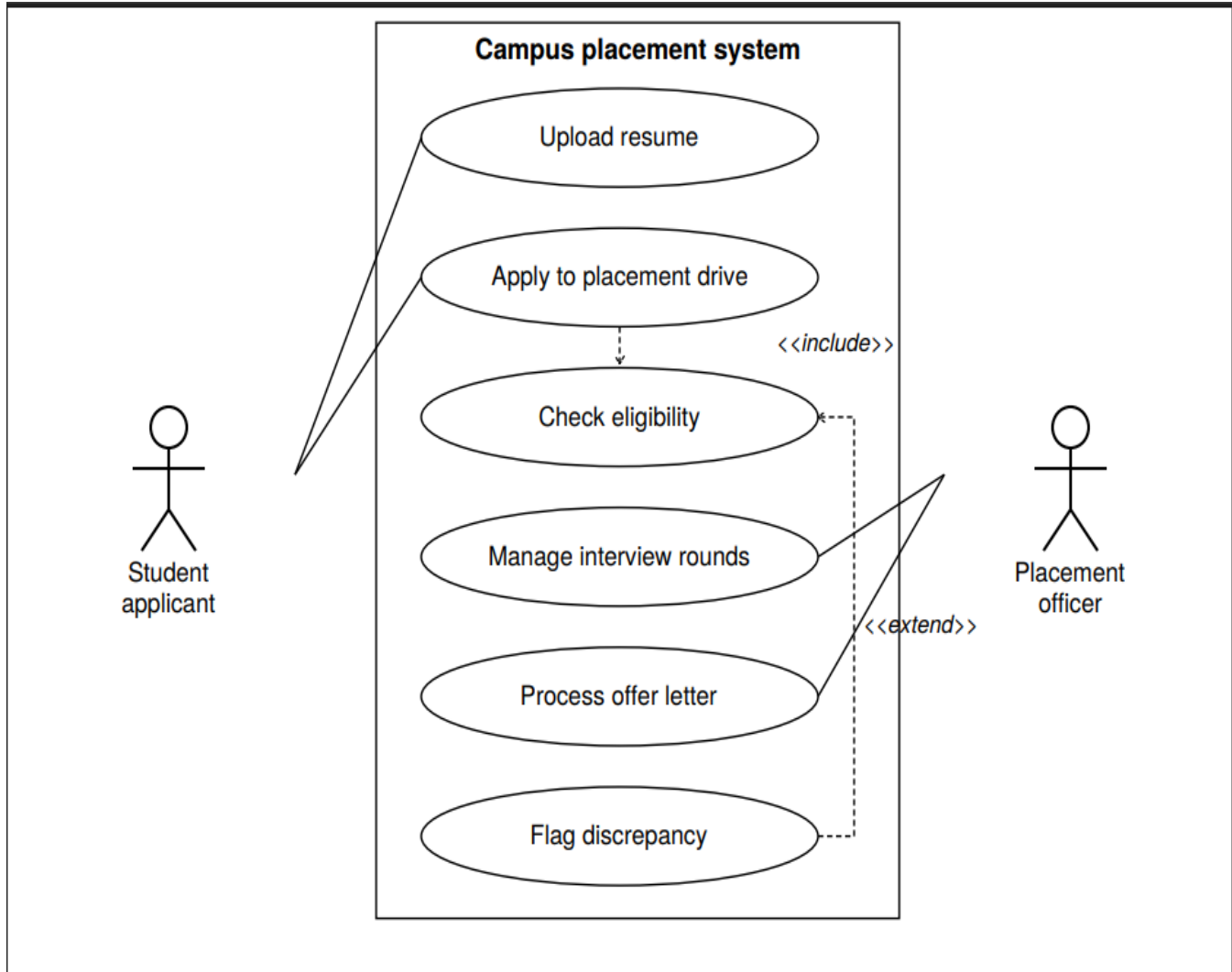
## Problem Statement 03: Campus Placement & Internship Pipeline

### 1. Requirements Table

| ID      | Type          | Description                                                                                                                                                   | Priority | Acceptance Criteria                                                                                                                                               | Rationale                                                                          |
|---------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| FR-001  | Functional    | The system shall parse uploaded student resumes (PDF format), extract skill tags, and match profiles against active company eligibility rules.                | High     | Pass: Ineligible applicants are filtered out with clear discrepancy reasons. Fail: Applicant with active backlog bypasses eligibility filter.                     | Core screening functionality for the entire placement pipeline.                    |
| FR-002  | Functional    | The system shall allow a student applicant to upload a resume and view the list of companies they are eligible to apply to.                                   | High     | Pass: After upload, the student sees only drives matching their CGPA/branch/backlog criteria. Fail: An ineligible drive appears in their list.                    | Gives students self-service visibility instead of manual checking by staff.        |
| FR-003  | Functional    | The system shall let a student applicant submit an application to a specific placement drive before its stated deadline.                                      | High     | Pass: Application submitted before deadline is recorded with a timestamp and confirmation. Fail: An application after the deadline is accepted.                   | Enforces fairness across all applicants for a drive.                               |
| FR-004  | Functional    | The system shall allow the placement officer to schedule and track candidates through multiple interview rounds (aptitude, technical, HR) for a given drive.  | Medium   | Pass: Officer can move a candidate from round 1 to round 2 and status updates instantly. Fail: A rejected candidate still appears in the next round's queue.      | Multi-round drives need a clear, auditable pipeline instead of spreadsheets.       |
| FR-005  | Functional    | The system shall allow the placement officer to generate and send an offer letter to a selected candidate, and record the candidate's accept/reject response. | High     | Pass: Offer letter is generated with correct candidate and company details and the response is logged. Fail: An offer is sent to a candidate not marked selected. | Formalizes the final, legally relevant step of the pipeline.                       |
| NFR-001 | Nonfunctional | The system shall encrypt all student academic records and offer documents at rest using AES-256.                                                              | High     | Pass: Bench marking tests confirm target latency and security standards under simulated peak load.                                                                | Academic and offer data are sensitive and must be protected.                       |
| NFR-002 | Nonfunctional | The system shall support at least 500 concurrent student applications during peak placement season without response times exceeding 3 seconds.                | Medium   | Pass: Load test with 500 simulated users shows 95% of requests under 3 seconds. Fail: Response times degrade past 3 seconds under that load.                      | Placement drives cause sudden traffic spikes; the system must not slow to a crawl. |

## 2. UML Use-Case Diagram

Actors: Student Applicant, Placement Officer. Includes one include relationship (Apply to Placement Drive → Check Eligibility) and one extend relationship (Flag Discrepancy → Check Eligibility).



### 3. Use-Case Flow Specification

#### Use Case: Check Eligibility

##### Preconditions:

Student has uploaded a resume and applied to a placement drive; company eligibility rules (CGPA cutoff, backlog limit, branch) are configured in the system.

##### Postconditions:

Applicant is marked either "Eligible" (application proceeds to the interview queue) or "Ineligible" (application is rejected with a reason).

##### Main Success Scenario:

- 1 Student submits an application to a placement drive.
- 2 System retrieves the student's academic record (CGPA, backlog count, branch).
- 3 System retrieves the company's eligibility rules for that drive.
- 4 System compares the student's record against the rules.
- 5 System marks the applicant "Eligible" and adds them to the interview queue.
- 6 System sends a confirmation notification to the student.

##### Alternate Flow – 4a. Applicant Fails Eligibility Check:

- a System marks the applicant "Ineligible" and records the specific reason (e.g. "CGPA below 7.5" or "Active backlog").
- b System notifies the student of the rejection and the reason.
- c Use case ends; applicant is not added to the interview queue.